



PREMIUM

Full-Length Mock Test

Mock 01

Electrical Engineering

GATE EE

Total Questions	60
Maximum Marks	108
Duration	180 minutes (3 Hours)
Exam Pattern	MCQ + MSQ + NAT
Negative Marking	MCQ/MSQ: -0.33 per wrong answer
Designed For	EduNeuro Premium

Easy 20%

Moderate 33%

Hard 47%

Based on Previous Year Question Pattern Analysis | Generated by EduNeuro Premium

General Instructions

This question paper contains THREE sections: Section A, Section B, and Section C.

Section A: Multiple Choice Questions (MCQ) — carries one or two marks each. Only ONE option is correct.

Section B: Multiple Select Questions (MSQ) — carries two marks each. ONE OR MORE than one option(s) is/are correct.

Section C: Numerical Answer Type (NAT) — carries one or two marks. Enter the numerical value in the space provided.

Negative Marking: -0.33 marks will be deducted for each wrong MCQ and each wrong MSQ answer.

No negative marking for NAT questions.

There are 60 questions for a total of 100 marks.

Duration: 3 hours (180 minutes).

Use the space provided for rough work. Scribbling on the question paper is permitted.

Do not ask for clarification from the invigilator. Interpret questions as given.

Paper Structure

Section	Type	Questions	Marks/Q	Total Marks	Negative
A	MCQ	35	1 / 2	35	-0.33
B	MSQ	5	2	10	-0.33
C	NAT	20	1 / 2	35	None
Total		60		100	

Marking Scheme

Correct MCQ: +1 mark (or +2 marks as specified per question)

Wrong MCQ: -0.33 marks

Correct MSQ: +2 marks

Wrong MSQ (partial or no match): -0.33 marks

Correct NAT: +1 mark (or +2 marks as specified per question)

No answer / Unattempted: 0 marks

NAT wrong answer: 0 marks (no negative marking)

Section A — Multiple Choice Questions (MCQ)

Q.1 [1 Mark] [EASY] [Electrical Machines — Induction Motors]

No-load current in transformer is called:

- A) Magnetizing current
- B) Excitation current
- C) No-load current
- D) All of the above

Q.2 [1 Mark] [EASY] [Power Systems — Protection]

Surge impedance loading (SIL) =:

- A) V^2/Z_c
- B) V/Z_c
- C) Z_c/V
- D) $V \times Z_c$

Q.3 [1 Mark] [EASY] [Power Electronics — Controlled Rectifiers]

Full-wave bridge rectifier PIV:

- A) V_m
- B) $2V_m$
- C) $V_m/2$
- D) $2 V_m$

Q.4 [1 Mark] [EASY] [Control Systems — State Space]

Ideal op-amp input impedance:

- A) Zero
- B) Infinite
- C) $1M\Omega$
- D) Output impedance

Q.5 [1 Mark] [EASY] [Signals & Systems — Fourier Analysis]

Slip in induction motor is:

- A) N_s/N_r
- B) $(N_s - N_r)/N_s$
- C) N_r/N_s
- D) $(N_r - N_s)/N_r$

Q.6 [1 Mark] [EASY] [Analog Electronics — Diodes]

DC machine with lap winding has brushes:

- A) 1
- B) P
- C) $P+1$
- D) 2

Q.7 [1 Mark] [EASY] [Electrical Machines — Servo Motors]

No-load current in transformer is called:

A) Magnetizing current

B) Excitation current

C) No-load current

D) All of the above

Q.8 [1 Mark] [EASY] [Power Systems — Load Flow]

Surge impedance loading (SIL) =:

- A) V^2/Z_c
- B) V/Z_c
- C) Z_c/V
- D) $V \times Z_c$

Q.9 [1 Mark] [EASY] [Power Electronics — Cycloconverters]

Full-wave bridge rectifier PIV:

- A) V_m
- B) $2V_m$
- C) $V_m/2$
- D) $\sqrt{2} V_m$

Q.10 [1 Mark] [EASY] [Control Systems — Compensators]

Ideal op-amp input impedance:

- A) Zero
- B) Infinite
- C) $1M\Omega$
- D) Output impedance

Q.11 [1 Mark] [EASY] [Signals & Systems — Convolution]

Slip in induction motor is:

- A) N_s/N_r
- B) $(N_s - N_r)/N_s$
- C) N_r/N_s
- D) $(N_r - N_s)/N_r$

Q.12 [1 Mark] [EASY] [Analog Electronics — Oscillators]

DC machine with lap winding has brushes:

- A) 1
- B) P
- C) $P+1$
- D) 2

Q.13 [2 Marks] [MODERATE] [Electrical Machines — Induction Motors]

No-load current in transformer is called:

- A) Magnetizing current
- B) Excitation current
- C) No-load current
- D) All of the above

Q.14 [2 Marks] [MODERATE] [Power Systems — Load Flow]

Surge impedance loading (SIL) =:

- A) V^2/Z_c

B) V/Zc

C) Zc/V

D) $V \times Z_c$

Q.15 [2 Marks] [MODERATE] [Power Electronics — Choppers]

Full-wave bridge rectifier PIV:

- A) V_m
- B) $2V_m$
- C) $V_m/2$
- D) $\sqrt{2} V_m$

Q.16 [2 Marks] [MODERATE] [Control Systems — Block Diagrams]

Ideal op-amp input impedance:

- A) Zero
- B) Infinite
- C) $1M\Omega$
- D) Output impedance

Q.17 [2 Marks] [MODERATE] [Signals & Systems — Convolution]

Slip in induction motor is:

- A) N_s/N_r
- B) $(N_s - N_r)/N_s$
- C) N_r/N_s
- D) $(N_r - N_s)/N_r$

Q.18 [2 Marks] [MODERATE] [Analog Electronics — Diodes]

DC machine with lap winding has brushes:

- A) 1
- B) P
- C) $P+1$
- D) 2

Q.19 [2 Marks] [MODERATE] [Electrical Machines — Induction Motors]

No-load current in transformer is called:

- A) Magnetizing current
- B) Excitation current
- C) No-load current
- D) All of the above

Q.20 [2 Marks] [MODERATE] [Power Systems — Protection]

Surge impedance loading (SIL) =:

- A) V^2/Z_c
- B) V/Z_c
- C) Z_c/V
- D) $V \times Z_c$

Q.21 [2 Marks] [MODERATE] [Power Electronics — Inverters]

Full-wave bridge rectifier PIV:

- A) V_m

B) $2V_m$

C) $V_m/2$

D) " 2 V_m

Q.22 [2 Marks] [MODERATE] [Control Systems — Block Diagrams]

Ideal op-amp input impedance:

- A) Zero
- B) Infinite
- C) $1M\Omega$
- D) Output impedance

Q.23 [2 Marks] [MODERATE] [Signals & Systems — Filters]

Slip in induction motor is:

- A) N_s/N_r
- B) $(N_s - N_r)/N_s$
- C) N_r/N_s
- D) $(N_r - N_s)/N_r$

Q.24 [2 Marks] [MODERATE] [Analog Electronics — Power Amplifiers]

DC machine with lap winding has brushes:

- A) 1
- B) P
- C) $P+1$
- D) 2

Q.25 [2 Marks] [MODERATE] [Electrical Machines — DC Machines]

No-load current in transformer is called:

- A) Magnetizing current
- B) Excitation current
- C) No-load current
- D) All of the above

Q.26 [2 Marks] [MODERATE] [Power Systems — Fault Analysis]

Surge impedance loading (SIL) =:

- A) V^2/Z_c
- B) V/Z_c
- C) Z_c/V
- D) $V \times Z_c$

Q.27 [2 Marks] [MODERATE] [Power Electronics — Choppers]

Full-wave bridge rectifier PIV:

- A) V_m
- B) $2V_m$
- C) $V_m/2$
- D) $\sqrt{2} V_m$

Q.28 [2 Marks] [MODERATE] [Control Systems — State Space]

Ideal op-amp input impedance:

- A) Zero

B) Infinite

C) 1M:•

D) Output impedance

Q.29 [2 Marks] [MODERATE] [Signals & Systems — Fourier Analysis]

Slip in induction motor is:

- A) N_s/N_r
- B) $(N_s - N_r)/N_s$
- C) N_r/N_s
- D) $(N_r - N_s)/N_r$

Q.30 [2 Marks] [MODERATE] [Analog Electronics — Op-Amps]

DC machine with lap winding has brushes:

- A) 1
- B) P
- C) $P+1$
- D) 2

Q.31 [2 Marks] [MODERATE] [Electrical Machines — Synchronous Machines]

No-load current in transformer is called:

- A) Magnetizing current
- B) Excitation current
- C) No-load current
- D) All of the above

Q.32 [2 Marks] [MODERATE] [Power Systems — Transmission Lines]

Surge impedance loading (SIL) =:

- A) V^2/Z_c
- B) V/Z_c
- C) Z_c/V
- D) $V \times Z_c$

Q.33 [2 Marks] [HARD] [Power Electronics — Rectifiers]

Full-wave bridge rectifier PIV:

- A) V_m
- B) $2V_m$
- C) $V_m/2$
- D) $\sqrt{2} V_m$

Q.34 [2 Marks] [HARD] [Control Systems — State Space]

Ideal op-amp input impedance:

- A) Zero
- B) Infinite
- C) $1M\Omega$
- D) Output impedance

Q.35 [2 Marks] [HARD] [Signals & Systems — Laplace Transform]

Slip in induction motor is:

- A) N_s/N_r

B) $(N_s - N_r)/N_s$

C) Nr/Ns

D) $(N_r - N_s)/N_r$

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Section B — Multiple Select Questions (MSQ)

Q.36 [2 Marks] [HARD] [Analog Electronics — Op-Amps]

Which are rotating electrical machines?

- A) Transformer
 - B) Induction motor
 - C) DC motor
 - D) Sync generator
 - E) Alternator
-

Q.37 [2 Marks] [HARD] [Electrical Machines — Synchronous Machines]

Which are rotating electrical machines?

- A) Transformer
 - B) Induction motor
 - C) DC motor
 - D) Sync generator
 - E) Alternator
-

Q.38 [2 Marks] [HARD] [Power Systems — Protection]

Which are rotating electrical machines?

- A) Transformer
 - B) Induction motor
 - C) DC motor
 - D) Sync generator
 - E) Alternator
-

Q.39 [2 Marks] [HARD] [Power Electronics — Rectifiers]

Which are rotating electrical machines?

- A) Transformer
 - B) Induction motor
 - C) DC motor
 - D) Sync generator
 - E) Alternator
-

Q.40 [2 Marks] [HARD] [Control Systems — State Space]

Which are rotating electrical machines?

- A) Transformer
 - B) Induction motor
 - C) DC motor
 - D) Sync generator
 - E) Alternator
-

Section C — Numerical Answer Type (NAT)**Q.51 [2 Marks] [HARD] [Power Electronics — Inverters]**

Transformer 415/230V, 50kVA. Full-load primary current (A)?

Answer: _____

Q.52 [2 Marks] [HARD] [Control Systems — Compensators]

3-phase induction motor: 4 poles, 50Hz. Sync speed (rpm)?

Answer: _____

Q.53 [2 Marks] [HARD] [Signals & Systems — Fourier Analysis]

Buck converter $V_{in}=48V$, $D=0.5$. $V_{out} = ?$

Answer: _____

Q.54 [2 Marks] [HARD] [Analog Electronics — BJT/FET Amplifiers]

Load factor = Avg demand / Peak demand = 0.6. For 100kW peak, avg = ?

Answer: _____

Q.55 [2 Marks] [HARD] [Electrical Machines — Servo Motors]

DC generator: 4-pole, wave wound, $Z=500$ conductors. Parallel paths = ?

Answer: _____

Q.56 [2 Marks] [HARD] [Power Systems — Fault Analysis]

Transformer 415/230V, 50kVA. Full-load primary current (A)?

Answer: _____

Q.57 [2 Marks] [HARD] [Power Electronics — Controlled Rectifiers]

3-phase induction motor: 4 poles, 50Hz. Sync speed (rpm)?

Answer: _____

Q.58 [2 Marks] [HARD] [Control Systems — State Space]

Buck converter $V_{in}=48V$, $D=0.5$. $V_{out} = ?$

Answer: _____

Q.59 [2 Marks] [HARD] [Signals & Systems — Filters]

Load factor = Avg demand / Peak demand = 0.6. For 100kW peak, avg = ?

Answer: _____

Q.60 [2 Marks] [HARD] [Analog Electronics — Op-Amps]

DC generator: 4-pole, wave wound, $Z=500$ conductors. Parallel paths = ?

Answer: _____

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Answer Key

Verify your answers after completing the full paper.

Q.1:	D	Q.2:	A	Q.3:	B
Q.4:	B	Q.5:	B	Q.6:	B
Q.7:	D	Q.8:	A	Q.9:	B
Q.10:	B	Q.11:	B	Q.12:	B
Q.13:	D	Q.14:	A	Q.15:	B
Q.16:	B	Q.17:	B	Q.18:	B
Q.19:	D	Q.20:	A	Q.21:	B
Q.22:	B	Q.23:	B	Q.24:	B
Q.25:	D	Q.26:	A	Q.27:	B
Q.28:	B	Q.29:	B	Q.30:	B
Q.31:	D	Q.32:	A	Q.33:	B
Q.34:	B	Q.35:	B	Q.36:	BCDE
Q.37:	BCDE	Q.38:	BCDE	Q.39:	BCDE
Q.40:	BCDE	Q.41:	BCDE	Q.42:	BCDE
Q.43:	BCDE	Q.44:	BCDE	Q.45:	BCDE
Q.46:	BCDE	Q.47:	BCDE	Q.48:	BCDE
Q.49:	BCDE	Q.50:	BCDE	Q.51:	121
Q.52:	1500	Q.53:	24	Q.54:	60
Q.55:	2	Q.56:	121	Q.57:	1500
Q.58:	24	Q.59:	60	Q.60:	2

Detailed Solutions

Question 1 [Electrical Machines — Induction Motors]

No-load current in transformer is called:

Answer: (D)

All terms refer to the same current at no-load condition.

Question 2 [Power Systems — Protection]

Surge impedance loading (SIL) =:

Answer: (A)

$SIL = V^2/Z_c$ where $Z_c = \sqrt{L/C}$ is surge impedance.

Question 3 [Power Electronics — Controlled Rectifiers]

Full-wave bridge rectifier PIV:

Answer: (B)

$PIV = 2V_m$ in full-wave bridge rectifier.

Question 4 [Control Systems — State Space]

Ideal op-amp input impedance:

Answer: (B)

Ideal op-amp has infinite input impedance.

Question 5 [Signals & Systems — Fourier Analysis]

Slip in induction motor is:

Answer: (B)

$s = (N_s - N_r)/N_s$. Fraction of synchronous speed lost.

Question 6 [Analog Electronics — Diodes]

DC machine with lap winding has brushes:

Answer: (B)

Lap winding needs P brushes for P poles.

Question 7 [Electrical Machines — Servo Motors]

No-load current in transformer is called:

Answer: (D)

All terms refer to the same current at no-load condition.

Question 8 [Power Systems — Load Flow]

Surge impedance loading (SIL) =:

Answer: (A)

$SIL = V^2/Z_c$ where $Z_c = \sqrt{L/C}$ is surge impedance.

Question 9 [Power Electronics — Cycloconverters]

Full-wave bridge rectifier PIV:

Answer:

(B)

$PIV = 2V_m$ in full-wave bridge rectifier.

Question 10 [Control Systems — Compensators]

Ideal op-amp input impedance:

Answer: (B)

Ideal op-amp has infinite input impedance.

Question 11 [Signals & Systems — Convolution]

Slip in induction motor is:

Answer: (B)

$s = (N_s - N_r)/N_s$. Fraction of synchronous speed lost.

Question 12 [Analog Electronics — Oscillators]

DC machine with lap winding has brushes:

Answer: (B)

Lap winding needs P brushes for P poles.

Question 13 [Electrical Machines — Induction Motors]

No-load current in transformer is called:

Answer: (D)

All terms refer to the same current at no-load condition.

Question 14 [Power Systems — Load Flow]

Surge impedance loading (SIL) =:

Answer: (A)

$SIL = V^2/Z_c$ where $Z_c = \sqrt{L/C}$ is surge impedance.

Question 15 [Power Electronics — Choppers]

Full-wave bridge rectifier PIV:

Answer: (B)

$PIV = 2V_m$ in full-wave bridge rectifier.

Question 16 [Control Systems — Block Diagrams]

Ideal op-amp input impedance:

Answer: (B)

Ideal op-amp has infinite input impedance.

Question 17 [Signals & Systems — Convolution]

Slip in induction motor is:

Answer: (B)

$s = (N_s - N_r)/N_s$. Fraction of synchronous speed lost.

Question 18 [Analog Electronics — Diodes]

DC machine with lap winding has brushes:

Answer:

(B)

Lap winding needs P brushes for P poles.

Question 19 [Electrical Machines — Induction Motors]

No-load current in transformer is called:

Answer: (D)

All terms refer to the same current at no-load condition.

Question 20 [Power Systems — Protection]

Surge impedance loading (SIL) =:

Answer: (A)

$SIL = V^2/Z_c$ where $Z_c = \sqrt{L/C}$ is surge impedance.

Question 21 [Power Electronics — Inverters]

Full-wave bridge rectifier PIV:

Answer: (B)

$PIV = 2V_m$ in full-wave bridge rectifier.

Question 22 [Control Systems — Block Diagrams]

Ideal op-amp input impedance:

Answer: (B)

Ideal op-amp has infinite input impedance.

Question 23 [Signals & Systems — Filters]

Slip in induction motor is:

Answer: (B)

$s = (N_s - N_r)/N_s$. Fraction of synchronous speed lost.

Question 24 [Analog Electronics — Power Amplifiers]

DC machine with lap winding has brushes:

Answer: (B)

Lap winding needs P brushes for P poles.

Question 25 [Electrical Machines — DC Machines]

No-load current in transformer is called:

Answer: (D)

All terms refer to the same current at no-load condition.

Question 26 [Power Systems — Fault Analysis]

Surge impedance loading (SIL) =:

Answer: (A)

$SIL = V^2/Z_c$ where $Z_c = \sqrt{L/C}$ is surge impedance.

Question 27 [Power Electronics — Choppers]

Full-wave bridge rectifier PIV:

Answer:

(B)

$PIV = 2V_m$ in full-wave bridge rectifier.

Question 28 [Control Systems — State Space]

Ideal op-amp input impedance:

Answer: (B)

Ideal op-amp has infinite input impedance.

Question 29 [Signals & Systems — Fourier Analysis]

Slip in induction motor is:

Answer: (B)

$s = (N_s - N_r)/N_s$. Fraction of synchronous speed lost.

Question 30 [Analog Electronics — Op-Amps]

DC machine with lap winding has brushes:

Answer: (B)

Lap winding needs P brushes for P poles.

Question 31 [Electrical Machines — Synchronous Machines]

No-load current in transformer is called:

Answer: (D)

All terms refer to the same current at no-load condition.

Question 32 [Power Systems — Transmission Lines]

Surge impedance loading (SIL) =:

Answer: (A)

$SIL = V^2/Z_c$ where $Z_c = \sqrt{L/C}$ is surge impedance.

Question 33 [Power Electronics — Rectifiers]

Full-wave bridge rectifier PIV:

Answer: (B)

$PIV = 2V_m$ in full-wave bridge rectifier.

Question 34 [Control Systems — State Space]

Ideal op-amp input impedance:

Answer: (B)

Ideal op-amp has infinite input impedance.

Question 35 [Signals & Systems — Laplace Transform]

Slip in induction motor is:

Answer: (B)

$s = (N_s - N_r)/N_s$. Fraction of synchronous speed lost.

Question 36 [Analog Electronics — Op-Amps]

Which are rotating electrical machines?

Answer:

(B, C, D, E)

Transformers are static. Others are rotating machines.

Question 37 [Electrical Machines — Synchronous Machines]

Which are rotating electrical machines?

Answer: (B, C, D, E)

Transformers are static. Others are rotating machines.

Question 38 [Power Systems — Protection]

Which are rotating electrical machines?

Answer: (B, C, D, E)

Transformers are static. Others are rotating machines.

Question 39 [Power Electronics — Rectifiers]

Which are rotating electrical machines?

Answer: (B, C, D, E)

Transformers are static. Others are rotating machines.

Question 40 [Control Systems — State Space]

Which are rotating electrical machines?

Answer: (B, C, D, E)

Transformers are static. Others are rotating machines.

Question 41 [Signals & Systems — Convolution]

Which are rotating electrical machines?

Answer: (B, C, D, E)

Transformers are static. Others are rotating machines.

Question 42 [Analog Electronics — Oscillators]

Which are rotating electrical machines?

Answer: (B, C, D, E)

Transformers are static. Others are rotating machines.

Question 43 [Electrical Machines — Servo Motors]

Which are rotating electrical machines?

Answer: (B, C, D, E)

Transformers are static. Others are rotating machines.

Question 44 [Power Systems — Economic Dispatch]

Which are rotating electrical machines?

Answer: (B, C, D, E)

Transformers are static. Others are rotating machines.

Question 45 [Power Electronics — Rectifiers]

Which are rotating electrical machines?

Answer:

(B, C, D, E)

Transformers are static. Others are rotating machines.

Question 46 [Control Systems — Root Locus]

Which are rotating electrical machines?

Answer: (B, C, D, E)

Transformers are static. Others are rotating machines.

Question 47 [Signals & Systems — Sampling]

Which are rotating electrical machines?

Answer: (B, C, D, E)

Transformers are static. Others are rotating machines.

Question 48 [Analog Electronics — Diodes]

Which are rotating electrical machines?

Answer: (B, C, D, E)

Transformers are static. Others are rotating machines.

Question 49 [Electrical Machines — Servo Motors]

Which are rotating electrical machines?

Answer: (B, C, D, E)

Transformers are static. Others are rotating machines.

Question 50 [Power Systems — Fault Analysis]

Which are rotating electrical machines?

Answer: (B, C, D, E)

Transformers are static. Others are rotating machines.

Question 51 [Power Electronics — Inverters]

Transformer 415/230V, 50kVA. Full-load primary current (A)?

Answer: (121)

$I = S/V = 50000/415 \approx 120.5 \approx 121 \text{ A}$.

Question 52 [Control Systems — Compensators]

3-phase induction motor: 4 poles, 50Hz. Sync speed (rpm)?

Answer: (1500)

$N_s = 120f/P = 120 \times 50/4 = 1500 \text{ rpm}$.

Question 53 [Signals & Systems — Fourier Analysis]

Buck converter $V_{in}=48\text{V}$, $D=0.5$. $V_{out} = ?$

Answer: (24)

$V_{out} = D \times V_{in} = 0.5 \times 48 = 24\text{V}$.

Question 54 [Analog Electronics — BJT/FET Amplifiers]

Load factor = Avg demand / Peak demand = 0.6. For 100kW peak, avg = ?

Answer:

$$\text{Avg} = 0.6 \times 100 = 60 \text{ kW.}$$

Question 55 [Electrical Machines — Servo Motors]

DC generator: 4-pole, wave wound, $Z=500$ conductors. Parallel paths = ?

Answer: (2)

Wave winding always has 2 parallel paths regardless of poles.

Question 56 [Power Systems — Fault Analysis]

Transformer 415/230V, 50kVA. Full-load primary current (A)?

Answer: (121)

$$I = S/V = 50000/415 \text{ "H } 120.5 \text{ "H } 121 \text{ A.}$$

Question 57 [Power Electronics — Controlled Rectifiers]

3-phase induction motor: 4 poles, 50Hz. Sync speed (rpm)?

Answer: (1500)

$$N_s = 120f/P = 120 \times 50/4 = 1500 \text{ rpm.}$$

Question 58 [Control Systems — State Space]

Buck converter $V_{in}=48\text{V}$, $D=0.5$. V_{out} = ?

Answer: (24)

$$V_{out} = D \times V_{in} = 0.5 \times 48 = 24\text{V.}$$

Question 59 [Signals & Systems — Filters]

Load factor = Avg demand / Peak demand = 0.6. For 100kW peak, avg = ?

Answer: (60)

$$\text{Avg} = 0.6 \times 100 = 60 \text{ kW.}$$

Question 60 [Analog Electronics — Op-Amps]

DC generator: 4-pole, wave wound, $Z=500$ conductors. Parallel paths = ?

Answer: (2)

Wave winding always has 2 parallel paths regardless of poles.
